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Inverse matrices



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In fact, the shapes wouldn't allow it.
But for square matrices, the shapes allow it and it happens, if A has an inverse.
So give me some cases--
let's see, I hate to be negative here, but let's talk about the case with no inverse.
So this is-- these matrices are called invertible or nonsingular.
Those are the good ones.

When a square matrix A has the property that for every vector b , the system $Ax = b$ has a unique solution, then A has an inverse matrix.

Definition 12.1 The **inverse** of an $n \times n$ matrix A is another $n \times n$ matrix A^{-1} such that

$$AA^{-1} = I \quad \text{and} \quad A^{-1}A = I.$$

Suppose that A represents the linear transformation f . Then A^{-1} exists if and only if an inverse function f^{-1} exists; in that case, A^{-1} represents f^{-1} .

Problem 12.2 Does the rotation matrix $R := \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ have an inverse? If so, what is it?

Solution: The inverse linear transformation is rotation by $-\theta$, so

$$R^{-1} = \begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

(As a check, try multiplying R by this matrix, in either order.)

Problem 12.3 Does the projection matrix $A := \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ have an inverse? If so, what is it?

Solution: The associated linear transformation f is not a 1-to-1 correspondence, because it maps more than one vector to 0 (it maps the whole z -axis to 0). Thus f^{-1} does not exist, so A^{-1} does not exist.

Algorithm to find the inverse of a matrix

Let A be an $n \times n$ matrix.

1. Form the $n \times 2n$ augmented matrix $[A|I]$.
2. Convert to RREF; the result will be $[I|B]$ for some $n \times n$ matrix B .
3. Then $A^{-1} = B$.

This is a special case of solving a matrix equation $AX = B$ since A^{-1} is the solution to $AX = I$.

Algorithm in practice



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I'll carry A over here and just do that multiplication.

I-- you'll see I'll do it the old-fashioned way.

7 minus 6 is a 1.

21 minus 21 is a 0.

Minus 2 plus 2 is a 0.

Minus 6 plus 7 is a 1.

Check.

So that is the inverse.

That's the Gauss--Jordan idea.

Inverses concept check

1/1 point (graded)

If **A** and **B** are two $n \times n$ matrices, what is $(\mathbf{AB})^{-1}$? (Hint: Think of **A** and **B** as functions from $\mathbb{R}^n$ to $\mathbb{R}^n$.)

☐ $\mathbf{A}^{-1}\mathbf{B}^{-1}$

☐ $\mathbf{A}^{-1}\mathbf{B}$

☒ $\mathbf{B}^{-1}\mathbf{A}^{-1}$

☐ $\mathbf{B}^{-1}\mathbf{A}$

☐ $\mathbf{AB}^{-1}$

☐ $\mathbf{BA}^{-1}$.



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You have used 1 of 3 attempts

Worked examples: inverses



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You just have to basically subtract, replace row 1 by row 1 minus b over a times row 2 plus row 3.

Let me write that down.

Row 1 minus b over a times row 2 plus row 3.

You want that to go into your row 1.

These stay the same.

Minus 1 over a minus b, 1 over a minus b.

This is where I'm going to go to my notes

and tell you that these numbers are 1 over a minus b, 0,

minus b over a, a minus b.

That looks pretty awful.

But that is A inverse.

And we're done, except it looks terrible.

So let me just write it in a nicer way.

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